

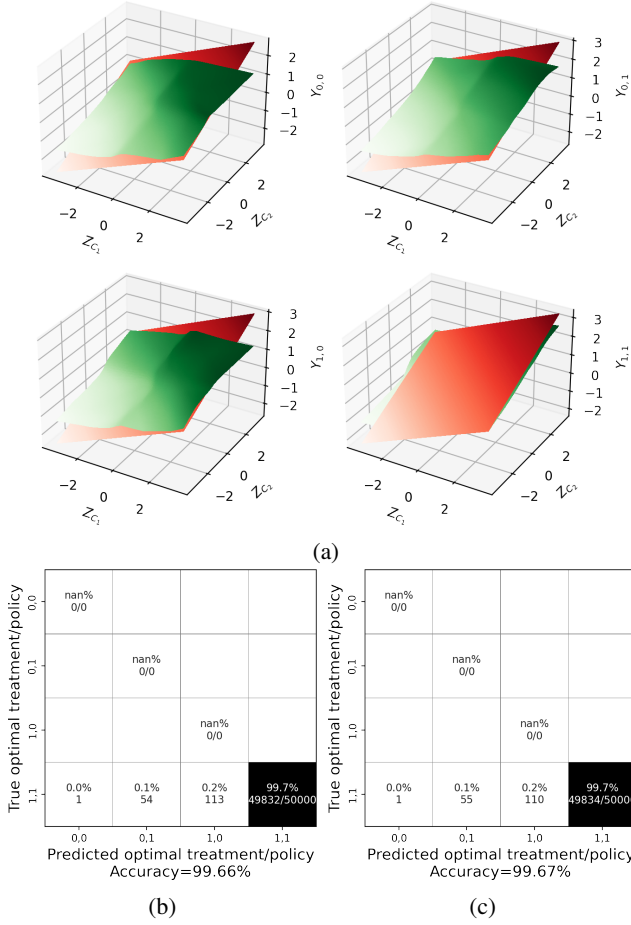


Figure 3: (a) The red and green surface-plots respectively represent the potential outcome maps of true SCM F and our *encapsulated-SCM* $\hat{F}$ generated from the SCM noise ranges $Z_{C1} \in [-3, 3]$, $Z_{C2} \in [-3, 3]$, $Z_Y = 0$. (b, c) Confusion matrices between the true and predicted optimal personalised policy on unseen validation and test sets respectively (with model under no misspecification setting from Fig. 2a).

hyperplanes in Fig. 3a (red). Observe that around the corners of the green 3D surface-plots which represents the low likelihood region of observing samples, the extrapolation assumption is slightly violated as the observational data corresponding to these regions are likely missing in the training set due to low likelihood. Extrapolation is an inevitable assumption made in almost all statistical models which is almost always violated under practical finite data setting. True extrapolation can be attained only with infinite data, which is impractical and unrealistic. In the technical appendix, we further present the same set of plots as in Fig. 3 under the complex experimental setting when both the models are misspecified and also the data exhibits effect modification/heterogeneity.

The confusion matrices from unseen validation and test sets in Figs. 3b and 3c indicate the true and predicted optimal individual-level treatment/policy $\bar{a}_1^\ell, \bar{a}_2^\ell$ for the ℓ^{th} indi-

vidual $\mathbf{X}^\ell$, which is predicted as below from Eq. (8)

$$\bar{a}_1^\ell, \bar{a}_2^\ell = \operatorname{argmax}_{a_1, a_2} Y_{a_1, a_2}(T\mathbf{X}^\ell). \quad (9)$$

Using c-GNF, we observe that the optimal treatments are indeed predicted with a very high accuracy of 99.6% on unseen validation and test sets under no-misspecification and no effect modification/heterogeneity in data. Similarly, under both model misspecifications we obtain 91% accuracy even with our sub-optimal hyperparameters. Thus, c-GNF enables precise P³A targeting and articulation of the tailored treatments/policies in the social sciences, instead of relying on ‘one-size-fits-all’ sub-optimal treatments/policies.

Finally, our c-GNF not only model the encapsulation of the true SCM, but also offer exact density estimation which provides necessary tools for causal inference practitioners to identify other causal quantities such as the probabilities of causation (??), i.e., Probability of Necessity (PN), Probability of Sufficiency (PS), Probability of Necessity and Sufficiency (PNS), Probability of Disablement (PD), Probability of Enablement (PE), that are of utmost importance to policy-makers planning various social programs.

Conclusion

This article developed causal-Graphical Normalizing Flow (*c-GNF*) for personalized public policy analysis (P³A). We demonstrated that our c-GNF learnt using only observational data without any assumptions on the exogenous noises or the functional mechanisms of the underlying SCM. We further identified c-GNF is on/above-par well-established standard causal effect estimation methods such as IPW and RWR. Most importantly, in contrast to IPW and RWR, using simulation experiments, we demonstrated the unique benefit of counterfactual inference using ‘*The First Law of Causal Inference*’ with c-GNF. Although our simulation study might be perceived as a limitation, the strong benchmarking of c-GNF on the simulated dataset demonstrated the usability of our framework with real-world data. The counterfactual inference demonstrated with c-GNFs showed potential to identify the optimal treatments at an individual-level, enabling precise P³A. When successful, P³A will likely accelerate the use of individualized policies beyond the closed settings of medical applications and into the open settings of the social science.

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Ethics Statement

The authors would like to indicate that the current work does not involve any real ethical considerations as assumptions are clearly stated and the experimental results with

benchmarking are based only on the non-sensitive simulated dataset.

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